

OF-3 Report :

Simulating Supersonic Flow over a step

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1 Introduction

This report entails simulations of Mach 3 supersonic flow over a step as shown in Figure 1. We first take a walk through the equations that govern this flow such as a simplified *Navier-Stokes* equations which make up Euler's equations and Normal Shock and Isentropic Flow theory.

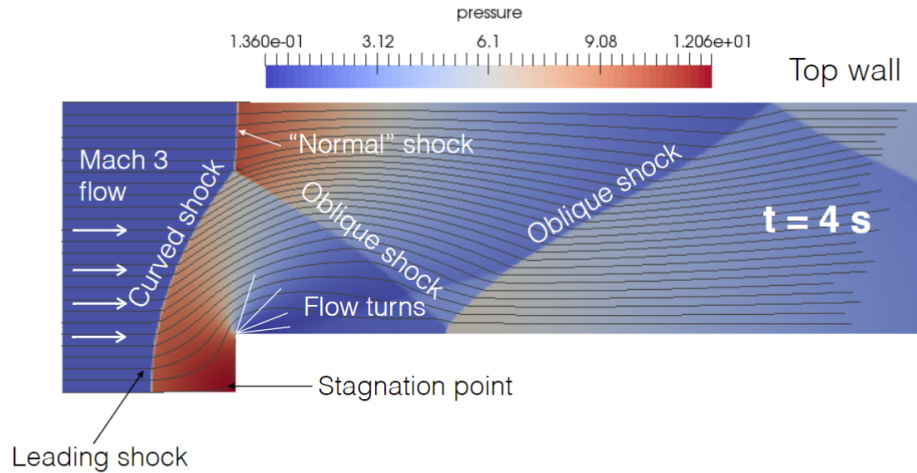


Figure 1: Details of the flow studied

We then simulate this flow using *OpenFoam's rhoCentralFoam* solver for various meshes from a coarse solution to more refined grids. *Paraview* is used to visualize the flow features such as the Mach number, velocity, temperature, pressure, and density. The stagnation point on the step is a point of interest as we compare the result of theory and CFD for the total pressure at this location. Lastly, we evaluate the performance of the parallel solver and observe how the number of processors affects the efficiency of the solver.

2 Governing Equations

2.1 Two-dimensional Euler equations

We consider the Euler equations in $\mathbb{R}^2$ complemented by the equation of state of an ideal gas (Eq. 6) with the specific heat ratio $\gamma = 1.4$. The two-dimensional Euler equations read:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} = 0 \quad (1)$$

Where $\mathbf{U}$ is the state vector, $\mathbf{F}$ is the x-flux vector, and $\mathbf{G}$ is the y-flux vector.

From the assumptions of Euler equations, e.g., no body forces and inviscid and isothermal flow, the simplified form *Navier-Stokes Equations* read:

- Continuity :

$$\frac{\partial \rho}{\partial t} + \rho \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0 \quad (2)$$

- X-Momentum :

$$\frac{\partial \rho u}{\partial t} + \rho \left(\frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y} \right) = -\frac{\partial p}{\partial x} \quad (3)$$

- Y-Momentum :

$$\frac{\partial \rho v}{\partial t} + \rho \left(\frac{\partial uv}{\partial x} + \frac{\partial v^2}{\partial y} \right) = -\frac{\partial p}{\partial y} \quad (4)$$

- Energy (inviscid simplification):

$$\frac{\partial E}{\partial t} + E \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = -p \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \quad (5)$$

Where $E = \rho(e + ||\mathbf{u}||^2/2)$. e is the internal energy of the fluid per unit mass (Joule per kilogram in SI).

Thus the state vector $\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho(e + \frac{1}{2}(u^2 + v^2)) \end{bmatrix} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ E \end{bmatrix}.$

The x-flux vector $\mathbf{F} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (\rho(e + \frac{1}{2}(u^2 + v^2)) + p) * u \end{bmatrix} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (E + p)(u) \end{bmatrix}$

The y-flux vector $\mathbf{G} = \begin{bmatrix} \rho v \\ \rho vu \\ \rho v^2 + p \\ (\rho(e + \frac{1}{2}(u^2 + v^2)) + p)(v) \end{bmatrix} = \begin{bmatrix} \rho v \\ \rho vu \\ \rho v^2 + p \\ (E + p)(v) \end{bmatrix}$

The *Ideal Gas Law* is our equation of state that relates the variables, ρ which is our fluids density, p the local pressure, R the molecular gas constant, and T the local temperature.

$$p = \rho RT \quad (6)$$

Using the first law of Thermodynamics, p can be related to the specific internal energy e and the specific heat ratio γ by:

$$e = \frac{p}{\rho(\gamma - 1)} \quad (7)$$

These can be combined for a new expression that expresses p entirely in terms of the state vector $\mathbf{U}$, allowing the flux vectors $\mathbf{F}(\mathbf{U})$ and $\mathbf{G}(\mathbf{U})$ to be evaluated directly from conserved quantities.

$$p = (\gamma - 1) \left(E - \frac{1}{2} \rho (u^2 + v^2) \right) \quad (8)$$

2.2 Local Sound Speed and Mach Number

To analyze compressible flow behavior, we often need the local sound speed and Mach number.

Local Sound Speed:

The local speed of sound a in a compressible fluid is given by the equation:

$$a = \sqrt{\gamma RT} = \sqrt{\gamma \frac{p}{\rho}} \quad (9)$$

Local Mach Number:

The local Mach number M is defined as the ratio of the flow speed to the local speed of sound:

$$M = \frac{\sqrt{u^2 + v^2}}{a} = \frac{|\mathbf{U}|}{a} \quad (10)$$

where u and v are the flow velocity components in the x and y directions, respectively.

2.3 Normal Shock and Isentropic Flow Relations

Here we look at the equations for normal shocks and isentropic flows, i.e., adiabatic and reversible.

For a normal shock with upstream (region 1) Mach number M_1 and downstream (region 2) Mach number M_2 , the following standard jump conditions apply (for a perfect gas with ratio of specific heats γ):

- **Static Pressure Ratio:**

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1} (M_1^2 - 1). \quad (11)$$

where:

- p_1 : upstream (pre-shock) static pressure,

– p_2 : downstream (post-shock) static pressure.

• **Downstream Mach Number:**

$$M_2^2 = \frac{1 + \frac{\gamma-1}{2} M_1^2}{\gamma M_1^2 - \frac{\gamma-1}{2}}. \quad (12)$$

Once the flow is on either side of a shock (and assuming no further shocks/viscous effects in that region), it can be treated as isentropic. In an isentropic flow of a perfect gas:

• **Isentropic Pressure Relation:**

$$p_0 = p \cdot \left(1 + \frac{(\gamma-1)}{2} M_1^2\right)^{\frac{\gamma}{\gamma-1}} \quad (13)$$

where:

- p_0 : total (stagnation) pressure,
- p : static pressure at a given point in the flow,
- M : local Mach number at that point.

2.4 Step Example

Considering a flow over our step at $M = 3$ and $p = 1$ Pa, what is the pressure at the stagnation point ($x=0.6\text{m}$, $y=0\text{m}$ in Figure 2) on the surface of the step? We will employ the equations laid out above.

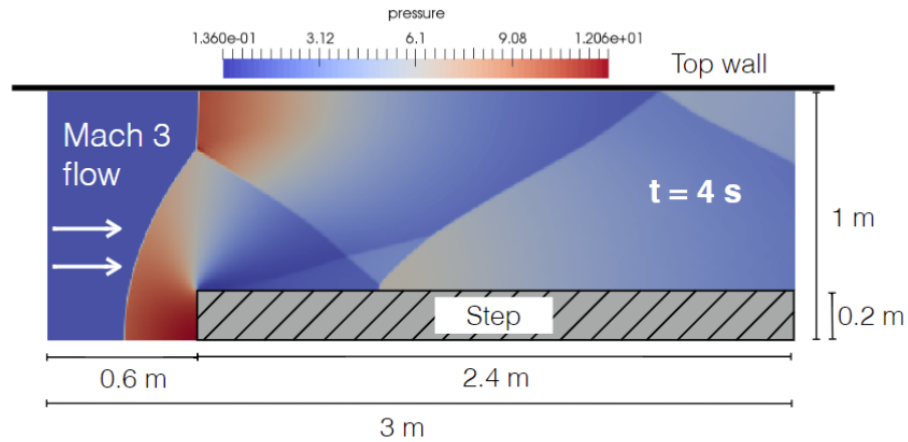
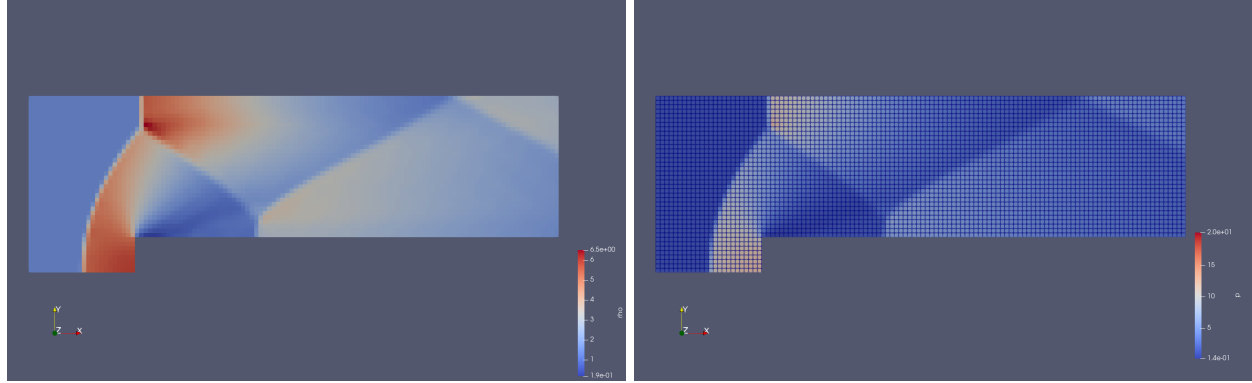


Figure 2: Overview of the Flow

Using Equation 13 we can obtain the total pressure in the first region $p_{0_1} = 36.73$ Pa. We can also obtain the local Mach past the shock using Equation 12, $M_2 = 0.4752$. We can obtain our static pressure past the shock as $p_2 = p_1 * \text{Eq. 11} = 10.33$ Pa. Thus using Equation 13 again, we get a stagnation pressure $p_{0_2} = 12.06$ Pa.

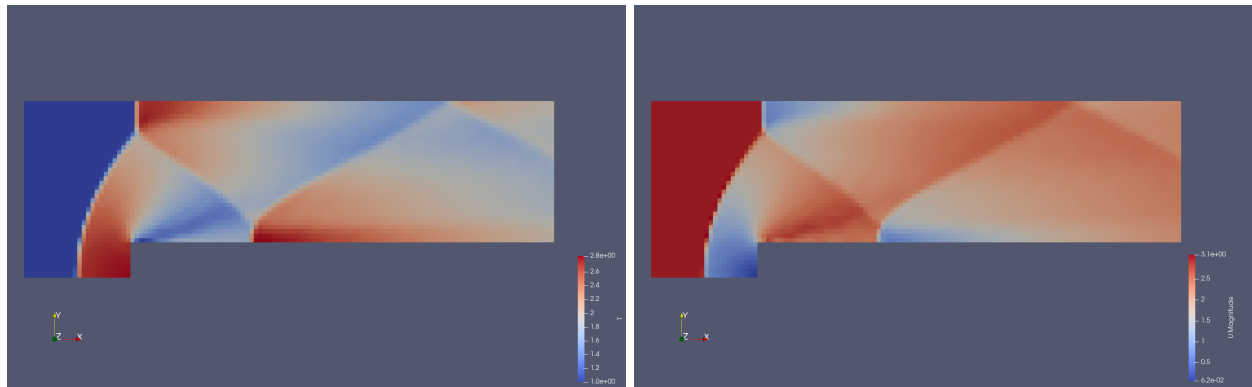
3 First Solution

In this section we visualize the properties pressure p , density ρ , velocity $\mathbf{U}$, and temperature T after simulating Mach 3 flow over the step for $T = 4$ s.



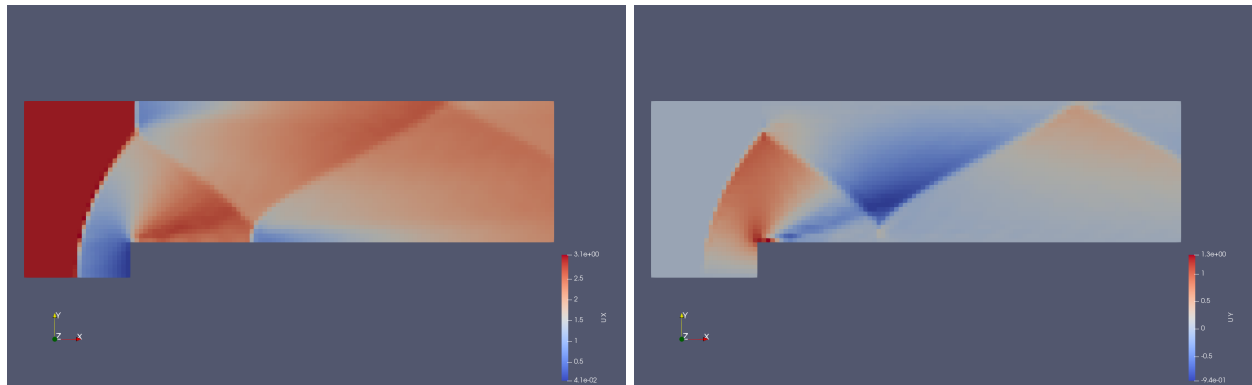
(a) Density ρ

(b) Pressure p



(c) Temperature T

(d) Magnitude of Velocity $|\mathbf{U}|$



(e) x-velocity u

(f) y-velocity v

Figure 3: Coarse mesh solution, gridpoints can be seen in the pressure plot

Using the built-in calculator in *Paraview*, we are able to employ Equations 9 and 10 to visualize the Mach and Speed of Sound in our supersonic simulation.

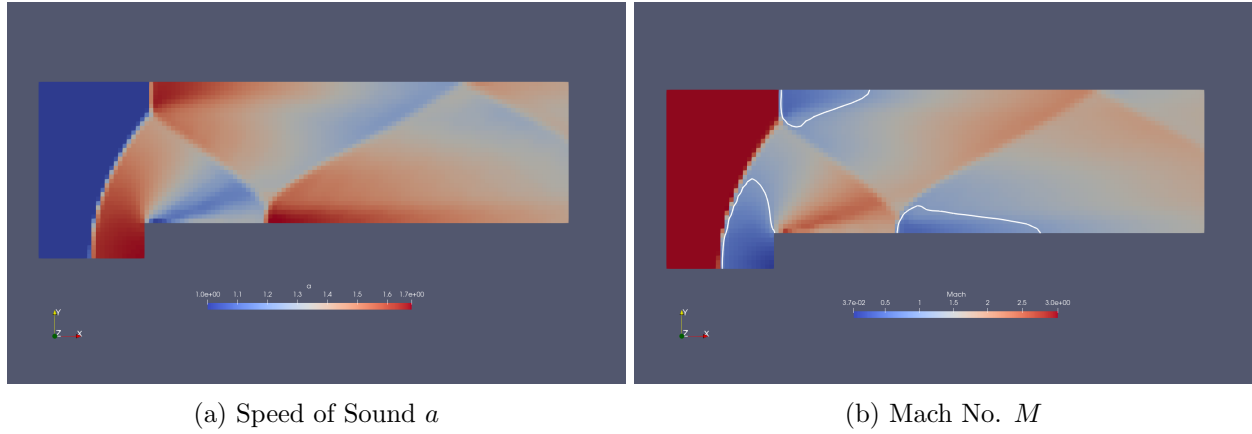


Figure 4: White countour lines representing sonic flow, i.e., where $M = 1$

The pressure at the stagnation grid point in this coarse mesh is $p = 11.9647$ Pa. This is close but slightly less than what was computed using our normal shock theory.

4 Converging the Solution

5 Parallel Solver Performance

In this section, we analyze the performance characteristics of the parallel version of the `rhoCentralFoam` solver. The primary objective is to evaluate how effectively the solver scales when computing one simulation time step on different numbers of processors. Ideally, as we increase the number of processors, $T(P)$, the time taken by the solver to advance one time step should decrease proportionally. We employ two critical performance metrics: parallel speedup and parallel efficiency. Parallel speedup is defined as:

$$\text{Speedup} = \frac{T(1)}{T(P)} \quad (14)$$

where $T(P)$ represents the computational time required by the solver advance the solution over one time step with P processors, and $T(1)$ is the baseline computational time. Parallel efficiency, η , is defined as:

$$\eta = \frac{T(1)}{T(P)} \quad (15)$$

indicating how effectively processor resources are being utilized.

We conducted scaling tests on a mesh of approximately one million cells, varying the number of processors from 1 up to 56 processors. The results of these scaling tests are summarized in Table 1 below, which lists the measured time step $T(P)$, calculated parallel speedup, and parallel efficiency for each tested processor P . Additionally, Figure 5 illustrates the observed speedup alongside ideal (linear) scaling and theoretical lines representing 80% and 60% parallel efficiency.

Processors (P)	Time $T(P)$ [s]	Speedup	Efficiency (η) [%]
1	1.4183	1.0000	100.00
2	0.6595	2.1505	107.53
4	0.3357	4.2239	105.60
8	0.1764	8.0392	100.49
16	0.1061	13.3668	85.54
32	0.0663	21.3712	66.79
56	0.0529	26.8098	47.87

Table 1: Parallel Performance Metrics Using Different # of Processors

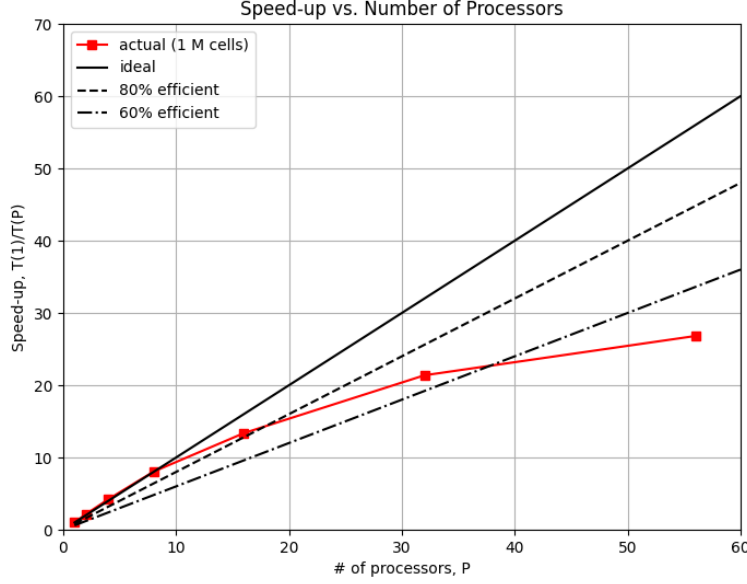


Figure 5: Speedup vs. Number of Processors P .

The collected data and graph clearly demonstrate that initially, as the number of processors increases, the speedup closely follows the ideal linear scaling, indicating efficient parallelization. This nearly linear behavior occurs because the computational workload is sufficiently large to ensure that each processor remains fully utilized, with minimal communication overhead and synchronization requirements. However, as the number of processors continues to increase beyond a certain point (around 16-32 processors in our tests), we observe diminishing returns in the speedup curve. This slowdown in parallel performance may occur due to several factors including: communication overhead, more processors need an increase in communication between them resulting in additional computational overhead, and synchronization delays, as processors counts rise, synchronizing the solver across multiple process introduces small yet accumulating errors.

This analysis highlights the importance of identifying an optimal number of processors to maximize computational efficiency. While adding processors initially leads to significant speed improvements, the overhead and inherent limitations eventually offset these gains. Thus, performance characterization, as demonstrated by our data, plays a crucial role in practical decision-making for deploying parallel computations effectively.

Appendix

A Code

PDF of code starts on next page.

Additionally, team repository: <https://github.com/andressuniaga/CFD-SP25-11>